AIORI HACKATHON 3 / PROBLEM STATEMENT PS-004

NEXORA

Resource-Aware Gateway-Assisted DNSSEC for Secure IoT Clients

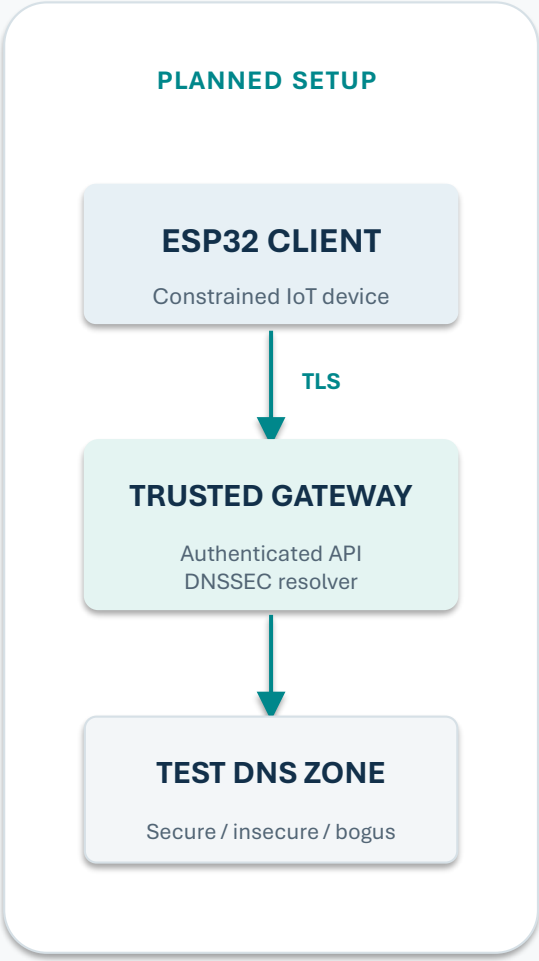
A practical prototype that moves DNSSEC validation from an ESP32 to a trusted local gateway.

TEAM

NEXORA

TEAM CODE

A3-PS004-TC412



Abstract & Problem Description

Abstract

Small IoT devices use DNS to find services. DNSSEC lets a validating resolver check signed DNS data, but full validation adds processing and memory work. NEXORA places validation at a trusted local gateway and protects the ESP32-to-gateway connection with TLS. We will test secure, insecure and bogus responses, verify that an unauthenticated gateway is rejected, and measure latency, client CPU time and memory use.

The pain point

- Without DNSSEC validation, a forged response can send an IoT device to the wrong service.
- Full cryptographic validation can add CPU and memory work on a small microcontroller.
- If validation moves to a gateway, the ESP32 must authenticate that gateway and trust its result.
- The tests must distinguish SECURE, INSECURE and BOGUS data and measure the added network and TLS overhead.

OUR APPROACH

Use a trusted gateway for DNSSEC validation. The ESP32 authenticates the gateway, and we measure the cost of this design.

Background

What each part does

DNS

Finds DNS records for a domain, including IP addresses.

DNSSEC

Checks the authenticity and integrity of DNS data; it does not encrypt traffic.

TLS

Lets the ESP32 verify the gateway and protects their communication.

DNSSEC outcomes used in the lab

SECURE	DNSSEC signatures and the chain validate.
INSECURE	No secure DNSSEC chain is established; apply the project policy.
BOGUS	Validation fails; do not use the answer.

In this design, DNSSEC validation runs at the gateway resolver, not on the ESP32. The ESP32 checks the gateway certificate before trusting its response. This design does not protect against a fully compromised gateway.

Literature & RFC Review

Resolver-assisted DNSSEC is an established approach. These standards and studies guide our design; our work is the ESP32 implementation, security tests and measurements.

SOURCE	WHAT THE SOURCE SAYS	HOW WE USE IT
RFC 9364 [1]	Summarizes DNSSEC standards and its use for authenticating DNS data; it does not encrypt queries.	Use DNSSEC for DNS data and TLS for the ESP32-to-gateway connection.
RFC 4035 [2]	Defines how signed DNS zones and validating resolvers handle DNSSEC data.	Use a controlled zone to test SECURE, INSECURE and BOGUS outcomes.
RFC 7858 / RFC 9846 [3,4]	RFC 7858 specifies DNS-over-TLS; RFC 9846 is the current TLS 1.3 standard.	Use TLS for the ESP32-gateway link; do not describe it as DNS-over-TLS.
IoT DNS Guidelines [5]	The 2026 Internet-Draft discusses resolver-assisted checking and full local validation for IoT devices.	Explain what the ESP32 trusts and keep the gateway inside the threat model.
CSProp, USENIX Security 2021 [6]	Proposes another way to reduce public-key verification costs, including DNSSEC on constrained devices.	It offers a different approach to compare; it does not establish novelty for our gateway design.

The gateway model itself is not new. NEXORA will contribute a working ESP32 implementation, security failure tests and repeatable measurements from our setup.

Proposed Solution

What we plan to build

01 **ESP32 IoT client**

Sends a domain request and checks the gateway TLS certificate.

02 **Gateway API on laptop**

Receives requests over TLS and asks the validating resolver for the DNS result.

03 **DNSSEC-validating resolver**

Checks DNSSEC against the test zone and reports the validation status.

04 **Test harness + logger**

Runs the DNS and gateway-authentication tests and records measurements.

How the client handles results

Gateway certificate check fails	STOP / REJECT
DNSSEC = SECURE	ACCEPT
DNSSEC = INSECURE	APPLY DECLARED POLICY
DNSSEC = BOGUS	REJECT

Initial setup: ESP32 Wi-Fi/TLS, a small gateway service on the laptop and a validating resolver such as Unbound. We will confirm compatibility during integration.

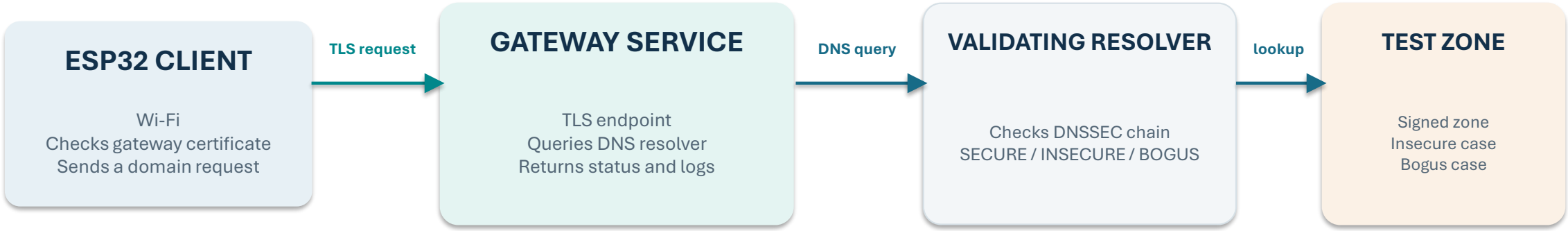
Solution Optimization

Moving validation to the gateway should reduce work on the ESP32, but it adds a network round trip and TLS overhead. We will measure both.

AREA	CURRENT PAIN POINT	NEXORA APPROACH
DNSSEC work	Full validation can add CPU and memory work on a small client.	The resolver validates DNSSEC; the ESP32 handles the request and status.
Channel trust	The client must trust the gateway and the path carrying its result.	The ESP32 checks the gateway certificate before sending a request.
Failure handling	A basic DNS lookup may not make the validation result clear to the client.	Record SECURE, INSECURE and BOGUS separately, along with gateway authentication.
Evaluation	Without controlled tests, we cannot claim a performance improvement.	Use a repeatable test zone and log latency, client CPU time and free heap.

Gateway assistance may reduce ESP32 work, but adds network and TLS overhead and makes the gateway a trusted dependency. Our tests will show the trade-off on this setup.

Solution Architecture & Design



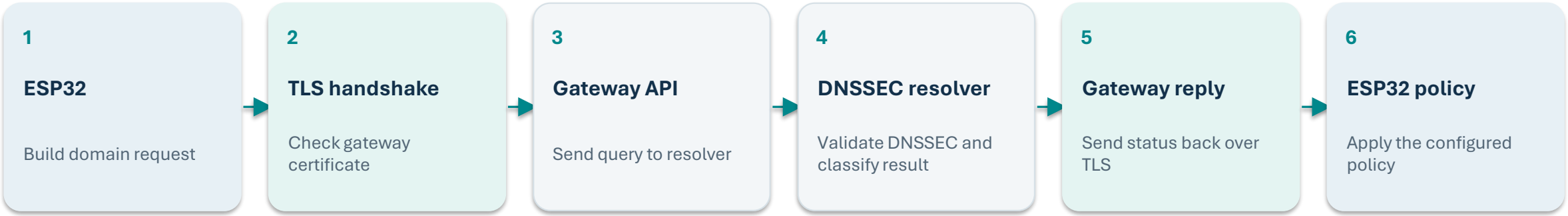
The resolver returns its validation status to the gateway. The gateway sends that status to the ESP32 over the same authenticated TLS connection.

What the client trusts

The ESP32 checks the gateway certificate. DNSSEC validation runs at the resolver, not on the ESP32. A compromised trusted gateway remains outside our protection.

We still need to complete integration, configure certificates and measure performance; this diagram shows the proposed setup.

Data Flow Diagram



Response policy

SECURE	Accept
INSECURE	Apply project policy; reject for protected test domains.
BOGUS	Reject
TLS / gateway auth failure	Do not use a DNS result

Deployment Timeline

Four-day implementation plan

1 2 3 4

Foundation

- Repository + roles
- ESP32 serial / Wi-Fi
- Threat boundary draft

DNSSEC lab

- Controlled test zone
- Validating resolver
- Gateway query API

Secure integration

- ESP32-to-gateway TLS
- Secure / insecure / bogus
- Unauthenticated gateway test

Evidence + delivery

- Repeat tests + measure
- Graphs + final PDF
- GitHub + working demo

We will finish the required security tests and demo first. An OLED or dashboard is optional and will be skipped if it puts the core system at risk.

Bibliography

[1] **P. Hoffman, “DNS Security Extensions (DNSSEC),” RFC 9364, BCP 237, Feb. 2023.**

https://www.rfc-editor.org/rfc/rfc9364.html

[2] **R. Arends et al., “Protocol Modifications for the DNS Security Extensions,” RFC 4035, Mar. 2005 (updated by subsequent RFCs).**

https://www.rfc-editor.org/rfc/rfc4035.html

[3] Z. Hu et al., “Specification for DNS over Transport Layer Security (TLS),” RFC 7858, May 2016.

https://www.rfc-editor.org/rfc/rfc7858.html

[4] E. Rescorla, “The Transport Layer Security (TLS) Protocol Version 1.3,” RFC 9846, July 2026.

https://www.rfc-editor.org/rfc/rfc9846.html

[5] A. Mishra et al., “IoT DNS Security and Privacy Guidelines,” Internet-Draft draft-ietf-iotops-iot-dns-guidelines-04, July 2026. Work in progress.

https://datatracker.ietf.org/doc/draft-ietf-iotops-iot-dns-guidelines/

[6] F. Alharbi et al., “CSProp: Ciphertext and Signature Propagation Low-Overhead Public-Key Cryptosystem for IoT Environments,” USENIX Security Symposium, 2021, pp. 609–626.
https://www.usenix.org/conference/usenixsecurity21/presentation/alharbi

The RFCs describe published standards. The IoT guidance is still an Internet-Draft. NEXORA performance claims will come from our own tests.

Thank you.

A prototype for secure DNS lookups from an ESP32 through a trusted gateway.

WHAT WE WILL SHOW

- ESP32 checks the gateway certificate before sending a request
- Secure, insecure and bogus DNS test cases
- Unauthenticated gateway connections are rejected
- Latency, client CPU time and memory measured across repeated tests

DNSSEC STATUS

SECURE

INSECURE

BOGUS

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Implementation and benchmark results will be added after testing.